Transformations of the roof space of Warsaw tenement houses in terms of economic profitability and the issues of protecting the historic cultural landscape

Abstract
As the number of construction projects increases, space in the city is becoming more and more scarce and a commodity in short supply. Historic areas are often the most saturated urban fabric, where there is no space for new development. Additional housing volume can usually be obtained as superstructures of existing buildings. Due to erection of new buildings in the areas valuable in terms of culture, a city may be plundered and devastated.
The article answers the question of how contemporary superstructures, conversions and adaptations of roof spaces are made for utility purposes. Various ways to shape roofs, as well as the main conditions of heritage conservation and spatial features that limit the freedom of development are discussed in the article. The article is supplemented with a specification of how the attic architectural layout influences the value of properties located on the top floor.

Keywords:
Attic adaptation, tenement house, monument, cultural landscape, return on investment, superstructure, housing development,

Introduction
Although city skylines are constantly changing, the urban landscape is still dominated by residential buildings. One of the most significant features of residential buildings is their location. It is particularly desirable to live near a business centre or in spaces with a distinctive individual nature - genius loci. Paradoxically, genius loci often attracts investors, developers, to its own detriment. New spaces arise by erecting or extending buildings or by making superstructures on the existing ones. In multiple cases, the added volume violates the existing character of the space. It is the duty of contemporary generations to protect places that bear witness to the past centuries from being destroyed and swallowed up by the voracious modern age (Terlikowski, p. 522, 523).
Work with any historic building should begin by identifying the value it represents both as an individual facility and as part of a wider context. As a matter of fact, any intervention resulting in an increase in floor space involves not only major technical, internal changes, but it also involves changes to the volume and external form, and thus, it also involves changes to the urban space and to changes of how both the building itself and the genius loci of the area are perceived. The most common form of acquiring new space is superstructure. Nowadays, superstructures are being built not only on listed buildings, but also on those which are aspiring to be listed (increasingly appreciated modernist buildings). All of them are a specific cultural landscape. Using basic guidelines referring to the form of the building, the permissible scope of facade changes can be determined. This determines the scale of changes that can be made to the building. As a general rule, the greater the permissible scale of transformation, the more usable space can be obtained and thus, the greater the developer's profit.

1. Objective, Scope, Method
The goal of the article was to examine how the superstructure of a historical residential building can be designed and what spatial and economic effects it will bring.
The first part of the article is based on a literature survey to determine heritage conservator's guidelines. The research field covers such issues as: a viewing analysis of the urban interiors, urbanism in a heritage conservator's perspective, technical conditions for the use of attics,

aesthetics, valuation and heritage conservator's guidelines, the permitted range of interference in a monument.
The second part contains a study of comparators, selected on the basis of the conclusions of the literature survey. This is a case study, based on objects that are typical for each category resulting from the study in part one. Spatial gains for each solution are identified. The results of the study in the second part are supplemented with economic data, i.e. the prices of 1 m^2 of dwelling places from each solution and descriptions of the most important differences between the dwelling places analysed.

2. Conditions of Heritage Conservation

For each historic site, as well as for those that aspire to be ones, a valuation analysis should first be carried out in order to determine the values that require protection, as well as to determine their hierarchy. Each facility should be analysed on a multi-scale - both the facility itself in particular and the context in which it functions.
The former scale is even perceived naturally, while the latter is often a problem or is overlooked as unimportant.
Historic urban layouts, even if they are not listed in the register of historical monuments, are an image of a bygone time, a specific cultural landscape and the local community identifies itself with its genius loci. This is why any possible spatial transformations should be carried out in a deliberate manner, taking into account the values that a given development presents. If no prior valuation analysis is carried out or its effect is ignored, this will lead to uncontrolled transformations resulting in decreasing values, devastation and sometimes even total loss of these values (Bojarowicz, Żaboklicki, p. 38).
When we first look at a facility in the urban, contextual scale that is often perceived subconsciously, it is not based on individual details or detailed architectural solutions, but on the volumes, surfaces and leading elements present on the entire facade. These are influenced by the height of the building (number of floors), the tectonics of the facade and the layout of the cornices. Therefore, the most important elements to be preserved in the transformed building are the height of the crowning cornice, calculated from the street level, and the slope of the roofline (Bojarowicz, Żaboklicki, p. 38).

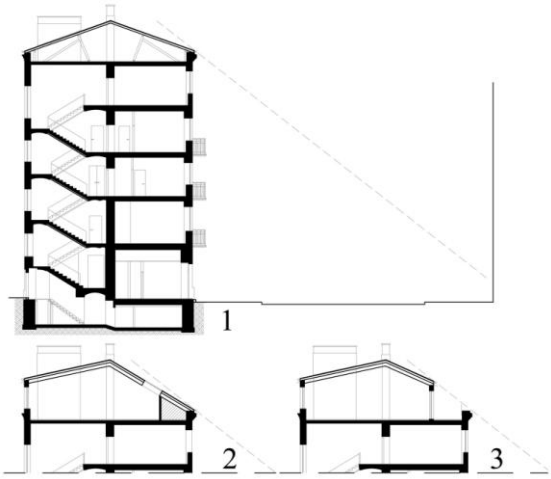
Attention should also be paid to the typical elements of a given building or style, so that by interfering with its form, it does not lose its distinctive features (Lewicki, s. 73).
Attention should be paid to the valuation analysis and abandoning the idea of equality of monuments in terms of how strict the differentiating rules of protection and transformation should be as the heritage conservator community suggests. A two-stage valorisation should be applied here:
a) 'identification' and 'location' of values in the structure of the monument taking into account their authenticity and the degree of integrity of the facility,
b) a comparative assessment of these values against the group of monuments appropriate in terms of typology that was separated in the national resources (Molski, p. 90).
At the same time, it should be remembered that the building is not taken out of context. Its form most often is in harmony or refers to the forms of surrounding buildings and creates a specific whole with them. Therefore, any interference with the facades of a single building should be viewed through the prism of this context.

3. Viewing Analysis

Viewing analyses should be the first step in assessing whether a building can be adapted. They will make it possible to determine two important issues:
a) option for transforming the roof so that it is invisible to a passer-by and therefore it does not interfere with how the building is perceived, and

b) option for transformations discernible, but made with respect for spatial order.
The first category depends on three elements: the width of the street, the height of the building and the observation point. They determine the angle of observation, and thus, the fragments that are discernible or beyond the observation capabilities of a passer-by.
Ways to adapt the building are listed below starting from the one that slightly interferes with the form of the building to the one that interferes most severely:
1. Maintaining the existing roofline angle using roof windows. This solution has a significant limitation: it is only feasible in buildings with high attics. The newly created interior will be characterised by an inefficient use of space at the facade, especially in facilities with low or no attic walls.
2. Shortening the roof plane and creating a terrace along the facade. This solution is seen as beneficial as it provides light to the new rooms with vertical windows and the use of (lost in the previous solution) space in the form of a terrace.
3. Changing the roofline angle so that it is imperceptible to the passer-by and creating adaptive conditions similar to item 1.
4. The use of asymmetrical rooflines by making the facade superstructure only on the courtyard side. A usable area is created at the façade and its transformation will not disturb the way the building is perceived by the user of the public space.
5. Use of gambrel roofs.
6. If the form of the building allows for more interference, a storey superstructure can be considered. Two solutions can be distinguished here:
a) partial superstructure on the courtyard side and superstructure of the outhouses to perceive the building the original way by the user of the public space,
b) overall superstructure with facade recomposition.

4. Case Study of Warsaw Townhouses
In order to illustrate the solutions detailed by the authors to obtain new usable floor space through the conversion or superstructure of historic residential buildings, the following examples from Warsaw will be presented. The buildings presented have the typical features of an inner-city historic buildings; they are several-storey tenement houses with front buildings and outhouses, and the transformations of the buildings were clear and vividly present the features described by the authors.



il. 1. Cross-section through the front townhouse at 33 Stalowa Street, Warsaw, Poland. Own elaboration.

Illustration 1 in Paragraph 1 shows the condition before revitalisation - the roof layout, typical of 20th/21st century Warsaw townhouses that in fact makes it impossible to convert the attic space into habitable rooms. Item 2. and Item 3. present transformation diagrams that make it possible to adapt the space analysed so that it would be invisible to a passer-by. The townhouse under analysis was modernised between 2018 and 2020 by the Praga-Północ Real Estate Management Board, so that it restores the original form. The ZGN did not decide to enlarge the residential space, e.g. by making superstructures on the outhouses, which are one storey lower than the front building, or adapting the attics.

Studies similar to those presented for the townhouse at 33 Stalowa Street are carried out for buildings that are the subject of developers' investments. For an investor, the most advantageous solution is the widest possible superstructure, but this is not always permissible. Then, different variants can be combined to preserve the appearance of the front elevation and, for example, to make superstructure of the front building from the courtyard side, including overall outhouse superstructure.

A good practice solution, which respects the value of the form of the original facade, is the building at 22 Jagiellońska Street in Warsaw, Poland, where only roof windows were used on the front side. The additional space was obtained by the investor through a complete superstructure of the outhouses and the facade from the side of the courtyard, together with replacement of the roof with an asymmetrical one (raising one roofline).

il. 2. Jagiellońska 22. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved January 28, 2023) with street view (https://www.google.pl/maps, 28.01.2023).

A similar solution was used in the redevelopment of the building at 16 Poznańska Street in Warsaw, Poland, where the angle of observation made it possible to obtain, in part, as many as two storeys in the roof space. In order to obtain even more usable space, superstructures were made on the outhouses. This superstructure also illustrates how the urban context - the adjacent buildings - can be taken into account. One of the neighbouring townhouses is one storey higher, which made it possible to make superstructure on the analysed townhouse by two storeys. The height to the level of the other neighbouring building was changed in a stepped way, in the form of a large terrace. Along the front elevation, the roofline was shortened, by placing a terrace, which at the same time made it possible to apply a greater angle of the roofline which was invisible to the user of the public space, but created a higher interior.

il. 3. Poznańska 16. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

With the use of a gambrel roof, an extra storey with high, attractive interiors - seemingly without superstructure, can be obtained. The building at 40 Smolna Street, where the shed roof, reconstructed after World War II contrary to the original form of the building, was replaced by the developer with a mansard roof, which probably crowned the building originally. The form and materials used clearly indicate that it is a contemporary element, what should be indicated as a model.

il. 4. Smolna 40. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

A different solution was suggested at 63 Al. Jerozolimskie, where the entire roof was raised by half a storey. Large dormer windows were used to illuminate the interiors with vertical

windows. The entire structure was covered with sheet metal, including the vertical walls, integrating them into the form of the rebuilt roof. It should also be noted that the height the roofline was raised refers to the plane of the roofline of the neighbouring building, facing it. This was facilitated by coordinating the superstructure of both buildings. The roofline slope is asymmetrical and on the courtyard side, the elevation was raised by a full storey, while its walls were consequently covered with sheet metal.

il. 5. Aleje Jerozolimskie 63. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

Considering the interference in how the public space looks like, the recomposition of the facade necessary for the superstructure of the front wall can be considered the most difficult (Fiuk, p. 308; Bizio, p. 124). The crowning cornice, which is a distinct element of composition, allows for a natural separation of the original part from the contemporary one. Incorporating parts that rise above the upper cornice, i.e. the corner turrets, often found in townhouses enclosing the frontage, is both necessary and may be a challenge for the architect. As an example of the composition discussed, we can mention the townhouse at 26 Stefana Okrzei Street in Warsaw, Poland, where the elevation of the front facade was designed using simple, contemporary forms referring to the historical part. It should be pointed out as unfavourable that due to the designed layout, the solid figure lost its vividness by absorbing elements rising above the cornice, and thus, the carefully restored and very impressive facade lost its expression.

il. 6. Stefana Okrzei 26. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

The issues raised do not occur in the case of buildings with poorer exposure, located in dense cluster of houses and with simple facade forms, such as at 1 Mała Street in Warsaw, Poland. In this townhouse superstructure, the new, higher wall was clad with sheet metal, which was also used to cover the roof, where with the original facade with its high attic wall, the superstructure elements could have been shortened.

il. 7. Mała 1. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

The townhouse at 19 Piękna Street in Warsaw, Poland, can be another example where the observation angles and slight setback of the extended facade in the form of a recess made it possible to almost hide the volume of the additional storey. The use of sheet metal cladding and glass elements emphasised its contemporary origin.

il. 8. Piękna 19. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

As the last example, the authors would like to point out a pretty controversial example, in terms of interference in the original appearance of the facade. It is a full-scale superstructure of a townhouse at 50 Hoża Street in Warsaw, Poland. Taking advantage of the neighbourhood of post-war buildings, the height of the street frontage was evened out by adding as many as two full storeys to the building discussed and using a flat roof. The raised part of the facade not only refers in terms of composition to the original part, but also duplicates the applied solutions. The crowning cornice is no longer an element that separates the ‘old from the new’, but has been relegated to a horizontal compositional detail of the façade. Its contemporary part has an enriched copy of it, which even more distorts the image of the whole. In this case, a person who is not an architect or an art historian that specialises in architecture of the period can not fully distinguish the contemporary interference.

il. 9. Hoża 50. Aerial View (source: https://ukosne.um.warszawa.pl, Retrieved 28.01.2023) with street view (https://www.google.pl/maps, 28.01.2023).

5. Analysis of the Impact of The Attic's Architectural Form on The Values of The Properties Located in Its Space

Based on recognition of the possible architectural conversion solutions of the last storey of townhouses, in relation to the existing conservation and urban planning conditions, analysis of the viability of each permitted new spatial arrangements can be made. The solution that allows to obtain the highest value of the newly created space is not always chosen by investors (Gawron, p. 60; Kozłowska, Rek-Lipczyńska, pp. 119-121), especially if a general refurbishment of the entire building is carried out. In such a case, all the residential units located in it need to be sold, whereas the way a given facility is perceived on the market by potential buyers has a very significant impact on the rate of flat sales. As a result of too much interference with the previously coherent historical fabric, the entire facility may be negatively perceived the city's residents, who are usually the target group for sales, and, as a consequence, they may not be willing to live in the building.

Buildings with a top floor being converted or with a superstructure have more options as to what spatial solutions may be chosen. If there is no need to sell many flats, less care can be taken when displaying new flats on the market. These flats are often bought by the building's residents.

In order to recognise how attractive particular architectural solutions are, buildings sold on the Warsaw primary market in 2017-2019 (the period before the pandemic that disrupted the real estate market) were analysed. In order to avoid significant differences in the value of the flats resulting from differences in location, only flats located in the Śródmieście district were accepted for the study.

The results are shown in the table below.

Solution Examined	Description	Price Range (rounded up to PLN 500 /m^2)
Building superstructure with facade face setback	Flats located in the superstructure are characterised by large areas (flats with at least 4 rooms). They are accompanied by additional space - terraces created by pulling back the facade face or additionally created terraces on the roofs of outhouses with no superstructures on them. Due to the location of the buildings analysed (Warsaw's Śródmieście District) and cultural conditions, access to additional spaces (balconies and terraces) is very high-priced.	PLN 17.000-25.000 /m^2
Building superstructure, new facade line flush with the facade of the other storeys	Flats located in the superstructure were characterised by diversified areas (flats with 2 to 4 rooms). The premises achieve lower prices, mostly do not have access to additional spaces (balconies and terraces). Balconies on the courtyard side are adjacent to individual premises. These are small areas balconies (approx. 2-3 m^2).	PLN 14.500-22.000 /m^2
A slight change in roof structure (implementation of roof windows or dormers)	The premises located in the roof space are characterised by the lowest standard (not all of the flats have a height of over 2.5 m). In addition, due to the sloping roofs, the flats are perceived as the least prestigious.	PLN 11.000-19.000 /m^2

6. Conclusions

More and more developments in cities and, in particular, in areas of dense, historical buildings, make investors who want to offer new housing convert and extend existing buildings. Prior to planning a development, it is essential to correctly identify the existing heritage conservation and urban planning conditions. It is necessary to analyse both the building that will be the subject of investment and its neighbourhood: the buildings that form the street frontage and the neighbouring public space.

Superstructures can be divided according to the spatial arrangements resulting from them: maintaining the existing roofline angle with the use of roof windows, shortening the roof plane and creating a terrace along the facade, changing the roofline angle at a degree invisible to passers-by, using asymmetrical rooflines by making superstructures on the facade only from the side of the courtyard, using gambrel roofs, superstructure on storeys. Care should be taken to ensure that the newly designed elements are not visible to users of the public space. Alternatively, to ensure that the newly added segments can be distinguished from the original fabric and, at the same time, not to dominate the historic form of the building.

Selection of a form that will be in harmony the existing historic buildings and will supplement it should be combined with an analysis of the value of the newly created space. However, preference should be given to architectural forms which will be accepted by the city's residents, since otherwise it may translate into difficult sales of flats.

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